PLE-SBS: A WAY DOWN FROM A FUELLED ROCKET, WITH THE POWER OFF, IN TWO MINUTES

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1 Where this came from

A roller coaster is a machine for giving people forces safely. In an earlier paper I designed coaster tracks under human g and jerk limits. The lesson that stuck: the thrill and the limit are the same number. The ride is good *because* it runs close to what a body can take, and safe because it never crosses that line.

A visit to ISRO turned the same idea toward crewed launch. In the hours before lift-off, astronauts and close-out technicians work 40–115 m up a tower beside a fully fuelled rocket. If something goes wrong they need a way down that beats the lifts, works with the power off and keeps working in a storm. NASA uses gravity-driven slidewire baskets, each engineered for one pad.

So this is a coaster designed for the worst day instead of the best one. Gravity drives it, a brake limits it, 1.5 g is the line it must not cross, and the only score is how fast the last person is behind a blast door.

The questions

- Can one **method** size a slidewire for *any* rocket on *any* pad?
- What makes the design **fail**, and what is left once the hardware works?
- Can every pod share **one wire** and still stop safely if a brake fails?

How I worked. The same loop each round: write down who needs what as requirements; design the physics on paper; build something that runs; run it until it breaks; put the failure back into the design. Five rounds of that loop are the spine of this poster.

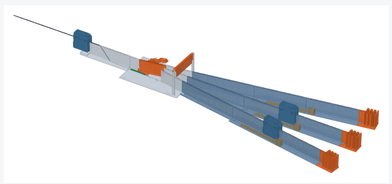
2 The problem, bounded

Scenario: alarm; crew unstrap and leave the capsule (60 s); walk the access arm; board; ride; leave the pod; reach shelter. The clock stops when the *last* person is protected. Worked example: 3 crew + 6 technicians on a low 42 m pad.

R1	Everyone protected within the window, at P95 not the average
R2	No electrical power: gravity drive, passive brakes
R3	Works in 17.5 m/s (34 kn) along-wire wind, either direction
R4	Peak deceleration ≤ 1.5 g, seated and belted
R5	No stalls, no pod within 5 m of another, dock cleared first
R6	Rope tension $\leq$ breaking strength / 5
R7	Terminus outside the fireball; shelter rated for the blast there

Five illustrative **danger levels** (my assumptions) set how much propellant joins an explosion and how long the warning is. Level 4, a pad fire, has a 120 s window, the Shuttle-era two-minute requirement.

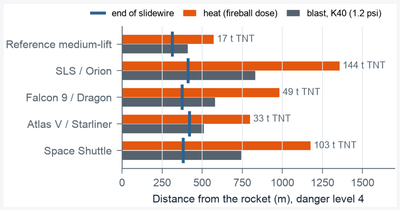
3 The 1:50 printed model



Only the 1.5 mm wire is bought; 26 printed parts make the rest, including block 8's switch yard in miniature. Each passing pod lifts a flap that rocks a two-finger escapement, and a spring steps the tongue to the next bay: 1, 2, 3 in turn. A printed comb, chain and bumper stop the pods.

4 Designing the physics

How far is far enough? Propellant becomes an equivalent mass of TNT (DoD rules; my own 10–20 % for hypergolics and methane). Blast distance scales as $R = 40\,W^{1/3}$ (1.2 psi). Fireball diameter and duration follow $D = 3.86M^{0.32}$ m, $t = 0.30M^{0.32}$ s, and a person is safe when the thermal dose $q^{4/3}t$ stays below $100\,(\mathrm{kW/m^2})^{4/3}$s. Reference launcher, level 4: 17.2 t TNT, blast 409 m, a 172 m fireball for 13 s, so **573 m** is required.



Heat sets the distance for every vehicle, and **every real slidewire ends inside it**. The wire is the first leg; a bunker or an armoured vehicle finishes the job.

The wire is not straight, and the pods are coupled. A taut string at constant tension sags under its own weight and under every pod. A pod behind *lifts* the wire under the pod ahead and can flatten its final approach until it stalls, so all pods on a wire are integrated together. Worked example: 36 mm rope at 150 kN sags 4.1 m; peak tension 154 kN against 181 kN allowed.

The ride and the brake. Each pod obeys

$$m\dot{v} = mg\sin\theta - \mu mg\cos\theta - \tfrac{1}{2}\rho C_d A\,(v - u\cos\theta)|v - u\cos\theta| - k_b v,$$

the last term a passive eddy-current brake. Its deceleration is $k_b v/m$, so **a light pod is braked harder than a heavy one**. I size k_b at two corners: a full pod in a 17.5 m/s tailwind must dock at ≤ 4 m/s; the lightest pod in a 17.5 m/s headwind must never drop below 1 m/s. When they conflict, the light pod wins and a catch net takes what the magnets cannot.

5 Building it, and checking it against NASA

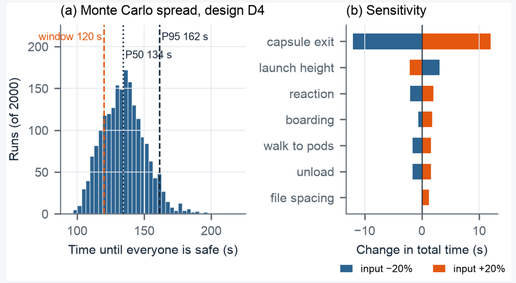
- **One physics file** (about 870 lines of dependency-free JavaScript) runs the website *and* produces every number and figure in the paper, so the paper cannot drift from the model.
- **Sixteen unit tests** against closed-form cases: $\sqrt{2gh}$ arrival, terminal speed, $wL^2/8T$ and $PL/4T$ sag, the chain-bed stop in v/ω, a stalled pod holding the one behind it. Halving the 0.02 s step moves the answer by 0.03 s.
- **Web simulator:** any rocket on any pad, dock or switch yard, net or chain bed, a brake you can fail. It animates the pods on the sagging wires with the hazard rings to scale and checks R1–R7. Works offline and on a phone.
- **Validation, no tuning.** Shuttle LC-39B (59 m drop, 362 m span): predicted peak 24.0 m/s against the reported "approximately 55 mph" (24.6 m/s), within 3 %. Artemis (114 m drop): gravity alone gives 77 mph against the reported 40–55 mph; a 300 m braked section holds it to 55 mph, so on a tall tower the magnets must act over much of the descent. Block 7 shows this is a general rule.

6 Running it until it broke: five redesigns

Each row is a full evacuation of 9 people at level 4, at both wind corners, and 1,000 times in Monte Carlo. Each problem found by *running* the system went back into the design.

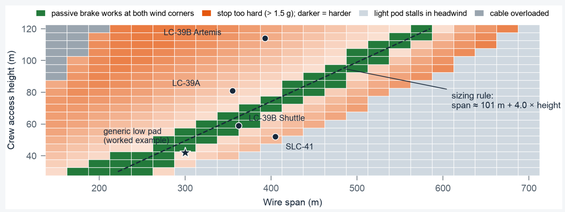
	Change	Time	P95	Failing	Worst stop
D0	Copy Artemis: 400 m, 1 wire × 3 pods in series, vehicle pick-up	203 s	231 s	93 %	16.4 g
D1	2 wires × 2 pods, balanced loading	184 s	213 s	91 %	16.4 g
D2	300 m span into a 3 m pit; light brake + 10 m catch net	188 s	218 s	0 %	1.4 g
D3	Bunker 40 m from the dock instead of a vehicle	160 s	192 s	0 %	1.4 g
D4	Crew pod at the arm exit, open pods, faster unload	**129 s**	**162 s**	0 %	1.4 g

D0 is the whole difficulty in one row. From only 42 m of height a 400 m wire ends almost flat. Any magnetic brake strong enough to matter strands a light pod in a headwind, so the auto-sizing chooses no brake, and a full pod in a tailwind hits the end at 25 m/s. Going parallel (D1) fixes nothing. **Geometry fixed it** (D2): shorter, steeper, into a pit, with a long-stroke net. D3 and D4 then attacked what was left, which was people.



Fourteen inputs varied together over 2,000 runs: median 134 s, P95 162 s, 3 runs flagged. D4 meets levels 1–3 at P95 but **not** the 120 s of level 4. The sensitivity says why: ±20 % on capsule exit moves the total by ±12 s, more than every hardware parameter combined. The crew reach the platform 78 s after the alarm, before any wire can help them.

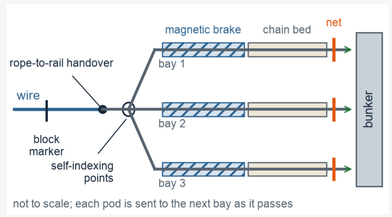
7 A sizing rule for any pad



The two-corner brake check for every crew height 30–120 m and span 150–700 m. Passive braking works only in a **narrow diagonal band**: left of it the stop exceeds 1.5 g, right of it a light pod stalls in a headwind. A line through the band gives $L_{\mathrm{span}} \approx 100\ \mathrm{m} + 4\,H_{\mathrm{crew}}$, a slope of about 1 in 7 to 1 in 5. Existing pads sit where their braking choices predict: Shuttle (passive nets and chains) on the band; LC-39A and Artemis on the steep side, where Artemis indeed brakes along the line; SLC-41 on the flat side, where riders control their own brakes. The band is 25–75 m wide, so the rule is a first cut and the simulator does the rest.

8 My add-on: one wire, a switch yard, a bay per pod

Existing systems give every basket its own wire and brake it at the end of that wire. In D4 that shared end is the bottleneck: each pod crawls through the brake zone, stops, unloads and is cleared before the next may arrive, a 37 s headway. D5 moves all of that **off the wire**.



- **Self-indexing points**, a turnout advanced by a ratchet each passing pod trips, send pod 1 to bay 1, pod 2 to bay 2. No power, no sensors.
- Each bay has a 60 m magnetic brake, a **drag-chain bed** (force grows with distance, not speed, so the stop is simple-harmonic) and a backup net, 15 m from the bunker door.
- **Release from inside** by a lever the occupants pull. A **block marker** on the wire, as in railway signalling, keeps it locked until the pod ahead is 60 m down; a stalled pod holds the next one at the platform.

What it buys

- Pods leave about **8 s apart** instead of 37 s.
- 24 people in six pods: **113 s** on one wire in a design headwind, against 286 s with a single dock and 111 s with six separate wires.
- A pod whose brake has **failed** stops at 1.40 g (1.76 g with a tailwind), where D4's catch net gave 3.8 g.
- Nine people: 131 s, P95 162 s, no failures in 1,000 runs, with one rope, one counterweight and one anchor.

9 What I learned

- **Design a safety system at its corners.** Nothing was decided by the average case, everything by the light pod in a headwind and the full pod in a tailwind.
- **The thrill and the limit are still the same number.** Passive braking is a band, not a knob: too little strands people, too much hurts them.
- **Running it is how you find out.** D0 looked like a sensible copy of a real system until it was operated: 93 % failure.
- **People, not pods, set the time.** Once the hardware worked, the remaining seconds were in the hatch, the arm and the boarding.

10 Honest limits and next steps

- No debris, basket sway or rope stretch. The launcher is a generic reference vehicle; danger levels and pad dimensions are my estimates; the hazard correlations are screening tools, not siting standards.
- One wire is one point of failure: a jammed trolley blocks everyone behind it, so crewed flights get a second wire as a backup. In the last minute before an explosion only the capsule's launch-escape system is fast enough.
- Next: timed runs on the printed model to fit μ and C_dA, model sway, test the rule on a real pad.